LECTURE 1.5 · SHIP AND OCEAN SYSTEMS DESIGN

The Design Equation

Papanikolaou (2026) §2.13, Appendix C · Session W1.4 · Lecture 20 min + Exercise 1.5 60 min

Tiago Santos, Luís Batista

Instituto Superior Técnico / University of Lisbon

tiago.santos@centec.tecnico.ulisboa.pt

luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

One Equation, Two Sides

Buoyancy: the hull

$$\Delta = \rho\, C_B\, L\, B\, T\, k_A \tag{1}$$

kA corrects the moulded volume for shell plating and appendages.

Weight: the ship and its load

$$\Delta = \mathrm{DWT} + W_{ST} + W_{OT} + W_M \tag{2}$$

Steel, outfit and machinery all grow with the ship, so Δ sits on both sides. Margins excluded here for simplicity.

In Exercise 1.4, Δ came from the deadweight and L, B, T from separate regressions. Put those L, B and T in the buoyancy side and you do not get back the same Δ: the two sides **miss by a few percent**. Today we choose the size so that both sides give the same Δ: that is the design equation.

Fix the Ratios, and One Dimension Is Left

Deadweight carriers: from Δ

Take L/B and B/T from similar ships and write L and T through B:

$$L = \left(\frac{L}{B}\right) B, \qquad T = \frac{B}{(B/T)}$$

Buoyancy then has one unknown, B:

$$\Delta = \rho\, C_B\, k_A\, \frac{(L/B)}{(B/T)}\, B^3$$

$$B = \left[\frac{\Delta\,(B/T)}{\rho\, C_B\,(L/B)\, k_A}\right]^{1/3} \quad (3) \qquad L = \left[\frac{\Delta\,(L/B)\,(L/T)}{\rho\, C_B\, k_A}\right]^{1/3} \quad (4)$$

And the draught directly, as in Exercise 1.5:

$$T = \left[\frac{\Delta}{\rho\, C_B\,(L/B)\,(B/T)^2\, k_A}\right]^{1/3} \quad (5)$$

Then B = (B/T)·T, L = (L/B)·B and D = (D/T)·T.

A · Before: Lecture 1.4

Table 1.8, with App. A fits

- Δ from DWT (Fig. A.51)
- L: App. A fit (A.53)
- C_B from Fn (Horn)
- B: App. A fit (A.54)
- T: App. A fit (A.56)
- D: App. A fit (A.55)
- Check: ρC_B LBT = Δ ? Ex. 1.4: −4.7 % → spiral

One at a time; buoyancy last
Book bases instead: L/$\nabla^{1/3}$, L/B, B/T, L/D

B · Today: design eq.

§1.3.7.1, step 2 by §2.13

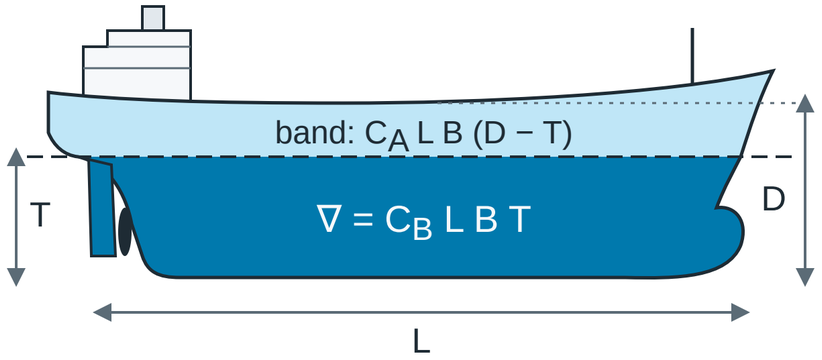
- Δ from DWT (Fig. A.51)
- L/B, B/T, D/T, C_B from similar ships
- T from Δ, eq. (5)
- B = (B/T)·T, L = (L/B)·B
  D = (D/T)·T
- Lightship from L, B, D
  Δ = W_L + DWT
- Check volume, T limit

Δ estimated first, then T;
B, L, D from the ratios;
↻ Δ confirmed by the weights (Exercise 1.5)

Hull Volume to the Deck: Where C_{BD} Comes From

A volume carrier fills its hull up to the deck, so we size it by that volume:

$$\nabla_D = C_{BD}\, L\, B\, D \quad (6)$$



Check · C_B 0.60, T/D 0.71

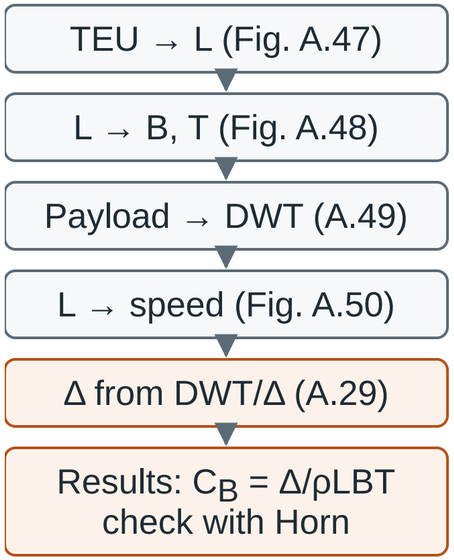
Wall-sided: $C_A = C_{WP} = (1 + 2C_B)/3 \rightarrow k = 1/3$, $C_1 = 0.24$ (book 0.25), $C_{BD} = 0.64$

Full box: $C_A = 1 \rightarrow k = 1$, $C_1 = T/D = 0.71$ (book up to 0.7), $C_{BD} = 0.71$

Split at the waterline	$C_{BD}\, LBD = C_B\, LBT + C_A\, LB\,(D - T)$
÷ L·B·D: an average weighted by height	$C_{BD} = C_B\, \frac{T}{D} + C_A\, \frac{D - T}{D}$
Name the band's share of the depth, h	$h = \frac{D - T}{D} = \frac{D}{D} - \frac{T}{D} = 1 - \frac{T}{D} \quad \Rightarrow \quad \frac{T}{D} = 1 - h$
Substitute T/D = 1 − h	$C_{BD} = C_B\,(1 - h) + C_A\, h$
Multiply out	$C_{BD} = C_B - C_B\, h + C_A\, h$
Regroup	$C_{BD} = C_B + (C_A - C_B)\, h$
C_A lies between C_B and 1 (a full box): place it a fraction k of the way	$C_A - C_B = k\,(1 - C_B), \qquad 0 \le k \le 1$
Substitute	$C_{BD} = C_B + k\, h\,(1 - C_B)$
Multiply by T/T = 1 to bring in (D − T)/T, as the book writes it	$h = \frac{D - T}{D} = \frac{D - T}{D} \cdot \frac{T}{T} = \frac{T}{D} \cdot \frac{D - T}{T}$
Result: equation (7), with $C_1 = k \cdot T/D$	$C_{BD} = C_B + \underbrace{k\, \frac{T}{D}}_{C_1}\, \frac{D - T}{T}\,(1 - C_B) \quad (7)$

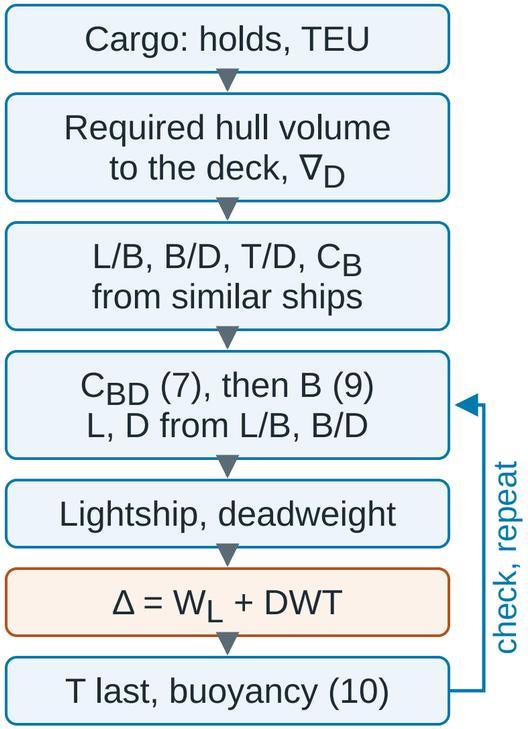
Volume Carriers: Size the Hull up to the Deck

A · Before: Lecture 1.4
Table 1.9, Kristensen curves



L, B, T read off curves;
C_B results, Horn checks it

B · Today: design eq.
§1.3.7.2, step 2 by §2.13



Volume first;
Δ and T come last

1	Start from the cargo, not Δ	Required hull volume ∇_D: holds (grain, bale, TEU), engine room, tanks (§2.17.2)
2	The hull volume as the box L·B·D times its fullness to the deck (slide 4); first T/D from similar ships	$\nabla_D = C_{BD}\, L\, B\, D$ (6) $C_{BD} = C_B + C_1 \dfrac{D-T}{T}(1-C_B)$ (7)
3	Fix L/B and B/D: L = (L/B)·B, D from B/D	$\nabla_D = C_{BD} \dfrac{(L/B)}{(B/D)} B^3$ (8)
4	One unknown, B (Eq. 2.103); then L and D from the ratios	$B = \left[\dfrac{\nabla_D (B/D)}{C_{BD} (L/B)} \right]^{1/3}$ (9)
5	Weights give Δ; the draught comes last (Table 1.9), then check T and C_{BD}	$T = \dfrac{\Delta}{\rho C_B L B k_A}$ (10)

The Reference Ship: Closing the Gap

Objective: find the main dimensions L, B, T, D of the reference ship for which buoyancy equals Δ = 46,050 t, keeping the ratios and CB of Exercise 1.4, and check T against the 11.5 m draught limit.

Brief (0.6, slide 2)	Exercise 1.4 (in class) · App. A.4 fits	Kept today
DWT 37,600 t · V 14.5 kn	Δ = 46,050 t (Fig. A.51)	L/B 5.92 · B/T 2.69 · D/T 1.36
T ≤ 11.5 m · ρ 1.025 t/m³ · kA = 1	L 174.7 · B 29.50 · T 10.97 · D 14.95 m	CB 0.757 (Horn, Fn 0.180)

1 The gap: buoyancy with those values

$$\rho C_B L B T = 1.025 \times 0.757 \times 174.7 \times 29.50 \times 10.97 = 43{,}880 \text{ t} \quad (-4.7\,\%)$$

2 Solve the design equation for T

$$T = \left[\frac{46{,}050}{1.025 \times 0.757 \times 5.92 \times 2.69^2} \right]^{1/3} = 11.15 \text{ m} \quad (5)$$

3 B, L and D from the ratios

$$B = 2.69 \times 11.15 = 29.99 \text{ m}, \quad L = 5.92 \times 29.99 = 177.5 \text{ m}, \quad D = 1.36 \times 11.15 = 15.16 \text{ m}$$

4 Check: both sides and the brief

$$\rho C_B L B T = 46{,}050 \text{ t} = \Delta, \qquad T = 11.15 \text{ m} < 11.5 \text{ m}$$

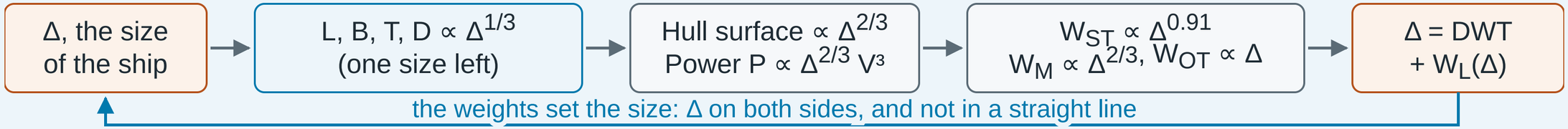
Is the gap really closed? No.

Buoyancy now matches Δ, but Δ is only a statistical guess (Fig. A.51): nobody has weighed this hull yet.

Next: each weight group as a power of Δ, and Δ = DWT + WL(Δ) solved by iteration.

Solving It: Iterate Until the Ship Floats

Why it is non-linear. The size sets the weights and the weights set the size: each weight group is a power of Δ, W = c·Δⁿ.



1
Write it as[1]

$$\Delta = g(\Delta)$$

$$g(\Delta) = \mathrm{DWT} + W_L(\Delta)$$

Start at Δ_0 = DWT, a ship with no lightship.

2
Iterate

$$\Delta_{k+1} = g(\Delta_k)$$

The weights of this ship give the next Δ.

3
Check

$$|g'(\Delta)| < 1$$

Each extra tonne of ship must cost less than a tonne of lightship.

The blue box: one size is left. With L/B, B/T, D/T and CB fixed, the shape is set and every dimension follows Δ (Exercise 1.5, Q2):

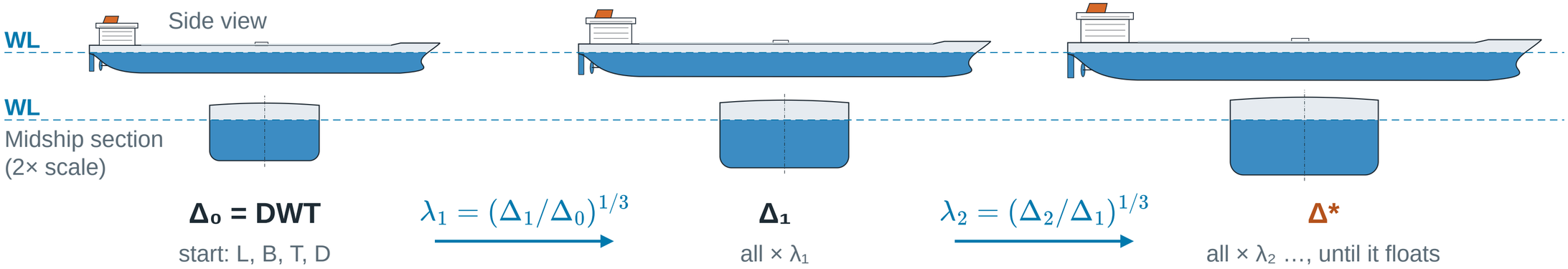
$$L = 4.953\,\Delta^{1/3}, \quad B = 0.837\,\Delta^{1/3}, \quad T = 0.311\,\Delta^{1/3}, \quad D = 1.36\,T$$

At every iteration the whole hull is rescaled to the new Δ. At least one dimension must follow Δ, (or the weights close but the hull does not float).

What follows Δ in the loop?	Δ (t)	Floats at Δ?
Nothing (L, B, T, CB from Ex. 1.4)	45,745	No: 4 % short
Only $T = \dfrac{\Delta}{\rho\, C_B\, L\, B}$	45,869	Yes, but B/T 2.57
All $T = \left[\dfrac{\Delta}{\rho\, C_B\, (L/B)\, (B/T)^2}\right]^{1/3}$	46,022	Yes, shape kept

[1] *Fixed-point iteration formulation.*

Same Shape, New Size at Every Iteration



1

The weights give a new Δ

$$\Delta_{k+1} = g(\Delta_k)$$

Steel, outfit and machinery of the current hull, plus DWT.

2

One scale factor

$$\lambda = \left(\frac{\Delta_{k+1}}{\Delta_k} \right)^{1/3}$$

The cube root, because displacement is a volume.

3

Every length grows alike

$$X_{k+1} = \lambda\, X_k, \quad X = L,\ B,\ T,\ D$$
$$\nabla_{k+1} = C_B\,(\lambda L_k)(\lambda B_k)(\lambda T_k) = \lambda^3\, \nabla_k$$

The same λ for every length: the ratios and the block coefficient are kept, and the hull floats at the new Δ.

Student activity

Review of numerical methods

Theory · Fixed-point iteration (successive substitution, Picard)

In words: the weight computed for the current Δ becomes the next Δ; stop when Δ stops changing (then buoyancy = weight).

$$f(x)=0 \iff x=g(x) \qquad x_{k+1}=g(x_k)$$
$$|g'(x)|<1 \text{ near } x^* \;\Rightarrow\; e_{k+1}\approx g'(x^*)\,e_k$$

Needs only g. Error × |g′(x*)| per pass: **linear**.

Theory · Newton–Raphson (tangent method)

In words: form the difference f = buoyancy − weight (the excess buoyancy) and drive it to zero, stepping along its tangent.

$$f(x)=0 \qquad x_{k+1}=x_k-\frac{f(x_k)}{f'(x_k)} \qquad f'(x^*)\neq 0 \;\Rightarrow\; e_{k+1}\approx \frac{f''(x^*)}{2f'(x^*)}\,e_k^2$$

Needs f and f′, a start close enough. Error squared: **quadratic**.

Example $\Delta=\underbrace{10\,000}_{\text{DWT}}+\underbrace{150\sqrt{\Delta}}_{\text{lightship}}$ Start at Δ_0 = DWT = 10 000 t · stop when the change is **below 0.01 t**

To compose the fixed-point iteration

1 · rearrange $\text{Buoyancy}=\text{weight}:\quad \Delta=10\,000+150\sqrt{\Delta}$
i.e. $f(\Delta)=0 \;\Leftrightarrow\; \Delta=g(\Delta)$

2 · g, g′ $g(\Delta)=10\,000+150\sqrt{\Delta} \quad g'(\Delta)=\dfrac{75}{\sqrt{\Delta}}\le 0.75<1$ ✓

3 · iterate $\Delta_{k+1}=10\,000+150\sqrt{\Delta_k}$

To compose Newton–Raphson

1 · f, f′ $f(\Delta)=\Delta-10\,000-150\sqrt{\Delta} \quad f'(\Delta)=1-\dfrac{75}{\sqrt{\Delta}}\ge 0.25\neq 0$ ✓

2 · iterate $\Delta_{k+1}=\Delta_k-\dfrac{\Delta_k-10\,000-150\sqrt{\Delta_k}}{1-75/\sqrt{\Delta_k}}$

Extended reading: where x = g(x) and the Newton step come from, derived step by step in Fixed Point and Newton–Raphson (two slides).

Student activity

Same root, 40 000 t: 17 evaluations of g against 5 Newton steps

Example $\Delta = \underbrace{10\,000}_{\text{DWT}} + \underbrace{150\sqrt{\Delta}}_{\text{lightship}}$ Start at Δ_0 = DWT = 10 000 t · count the evaluations of g until the change is **below 0.01 t**

k	Δ_k	$150\sqrt{\Delta_k}$	Δ_{k+1}	change	ratio
0	10 000.0	15 000.0	25 000.0	+15 000.0	
1	25 000.0	23 717.1	33 717.1	+8 717.1	0.581
2	33 717.1	27 543.3	37 543.3	+3 826.2	0.439
3	37 543.3	29 064.1	39 064.1	+1 520.8	0.397
⋮	⋮	⋮	⋮	⋮	⋮
14	39 999.95	29 999.98	39 999.98	+0.032	0.375
15	39 999.98	29 999.99	39 999.99	+0.012	0.375
16	39 999.99	30 000.00	40 000.00	+0.005 ✓	0.375

Fixed-point iteration: 17 evaluations of g (k = 0 to 16). Change × 0.375 = g′(Δ*) per pass: linear.

k	Δ_k	$f(\Delta_k)$	$f'(\Delta_k)$	Δ_{k+1}	change
0	10 000.0	−15 000.0	0.250	70 000.0	+60 000.0
1	70 000.0	+20 313.7	0.717	41 649.7	−28 350.3
2	41 649.7	+1 037.3	0.633	40 009.7	−1 640.0
3	40 009.7	+6.05	0.625	40 000.0	−9.68
4	40 000.0	+0.0002	0.625	40 000.0	−0.0004 ✓

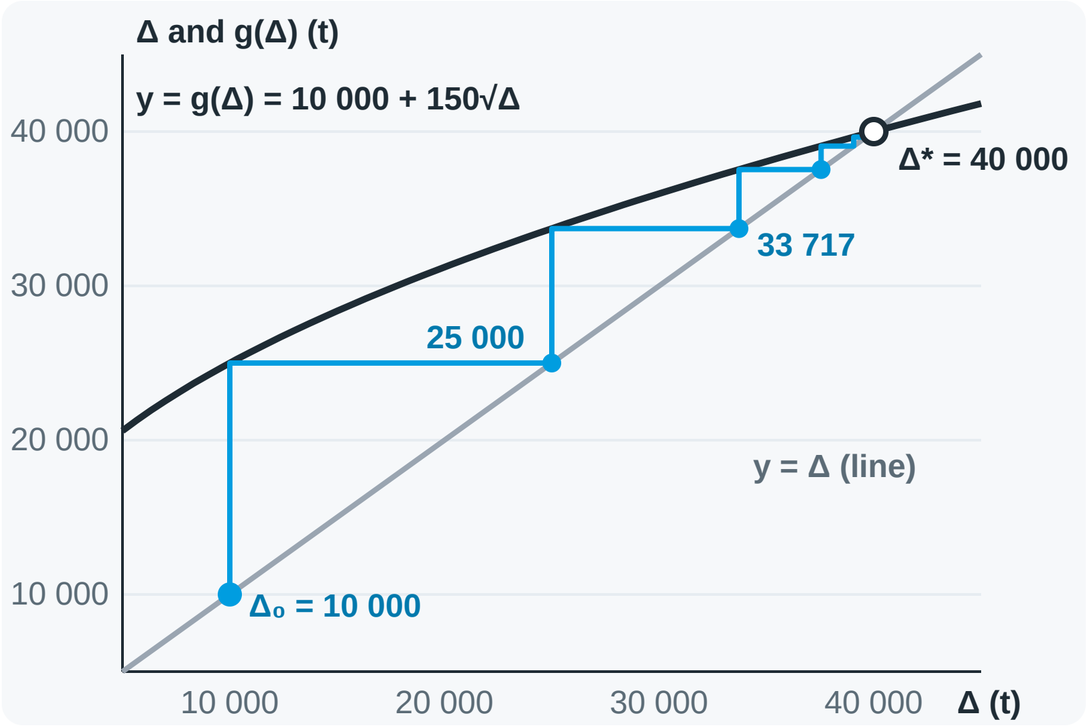
k = 0: 15 000 t short, slope only 0.25: grow by 15 000 / 0.25 = 60 000 t, far past Δ*.

Newton–Raphson: 5 steps (k = 0 to 4), each one evaluation of g and g′. Once close, error 1 650 → 9.7 → 0.0004 t: squared each step, quadratic.

Student activity

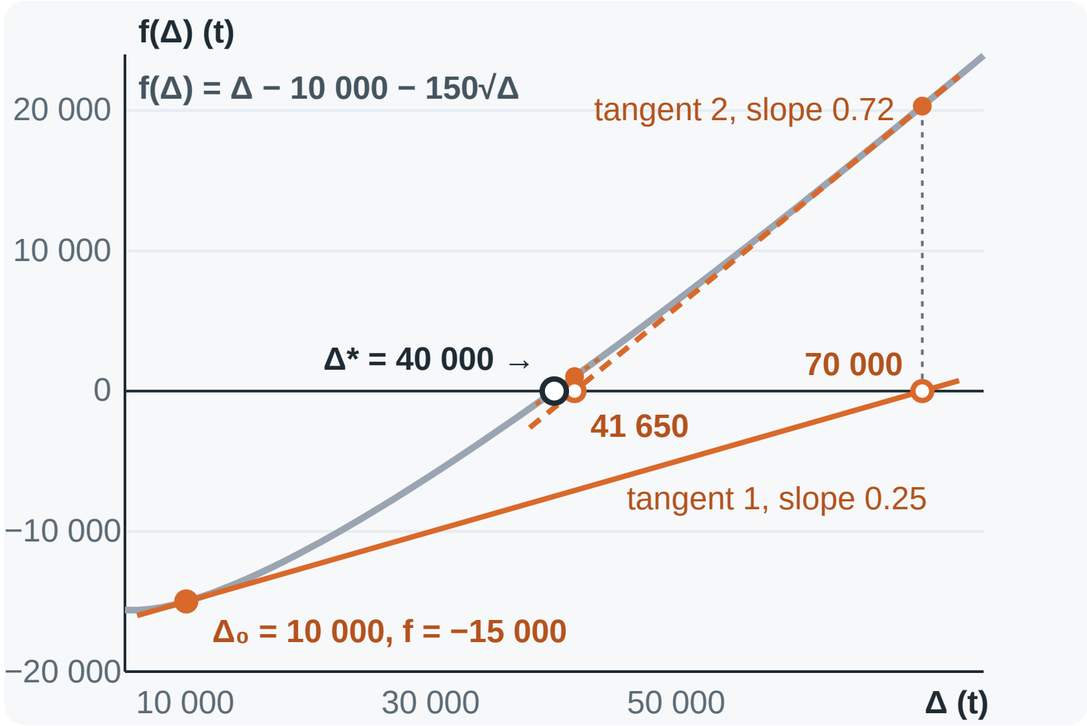
Fixed point climbs a staircase; Newton follows tangents

Fixed-point iteration · Δ = g(Δ)



Δ = g(Δ) = DWT + WLS(Δ): the equilibrium is “buoyancy = weight computed from the current size”.

Newton–Raphson · f(Δ) = 0



f(Δ) = buoyancy − weight: the equilibrium is “excess buoyancy = 0”.

Student activity

Sizing the reference tanker: 37 600 t deadweight

Objective: determine the displacement Δ and the main dimensions using simplified model equations and an iterative solution. The owner’s brief (0.6) fixes the deadweight and the service speed; the proportions and the block coefficient come from Exercise 1.4. The weight of the ship is assumed to break down as:

$$W = \mathrm{DWT} + \underbrace{W_{\mathrm{st}} + W_{\mathrm{o}} + W_{\mathrm{m}}}_{W_{\mathrm{LS}}}$$

Steel	$W_{\mathrm{st}} = K\,E^{1.36} \quad E = L\,(B + T) + 0.85\,L\,(D - T)$
Outfit	$W_{\mathrm{o}} = c_{\mathrm{o}}\,\Delta$
Machinery	$W_{\mathrm{m}} = k_{\mathrm{m}}\,P_{\mathrm{B}} \quad P_{\mathrm{B}} = \dfrac{\Delta^{2/3}\,V^{3}}{C_{\mathrm{adm}}}$

Item	Symbol	Value
Deadweight	DWT	37 600 t
Service speed	V	14.5 kn
Seawater density	ρ	1.025 t/m³
Block coefficient (Ex 1.4)	C_{B}	0.757
Length / beam (Ex 1.4)	L/B	5.92
Beam / draught (Ex 1.4)	B/T	2.69
Depth / draught (Ex 1.4)	D/T	1.36
Steel coefficient (tankers)	K	0.032
Outfit coefficient	c_{o}	0.025
Machinery weight	k_{m}	0.10 t/kW
Admiralty coefficient	C_{adm}	450

Student activity

Buoyancy = weight; Δ sets the dimensions

Q1 · Equilibrium equation

Write displacement from buoyancy computation = displacement from weight components.

$$\rho\, C_{\mathrm{B}}\, L\, B\, T = \mathrm{DWT} + W_{\mathrm{st}} + W_{\mathrm{o}} + W_{\mathrm{m}}$$

- Known: DWT = 37 600 t. Unknown: Δ.
- Lightship > 0, so Δ > DWT.
- Δ_0 = DWT (zero lightship) is a lower bound and a natural start.

Q2 · Displacement equation as function of non-dimensional ratios

Write Δ with L/B, B/T and the block coefficient; then T(Δ).

1 · Inject the ratios (in blue) into the displacement equation

$$B = (B/T)\,T \qquad L = (L/B)\,B = (L/B)\,(B/T)\,T$$

$$\Delta = \rho\, C_{\mathrm{B}}\, \underbrace{(L/B)\,(B/T)\,T}_{L}\, \underbrace{(B/T)\,T}_{B}\, T = \rho\, C_{\mathrm{B}}\, (L/B)\,(B/T)^2\, T^3$$

2 · Draught from Δ, then B, L and D = (D/T)·T

$$T = \left(\frac{\Delta}{\rho\, C_B\, (L/B)\,(B/T)^2} \right)^{1/3} = 0.3110\, \Delta^{1/3}$$

For a given Δ, with L/B and B/T fixed, L, B and T are determined.

3 · Similar equations follow with L or B as the variable

$$\Delta = \frac{\rho\, C_B\, L^3}{(L/B)^2\,(B/T)} \;\Rightarrow\; L = 4.953\, \Delta^{1/3} \qquad \Delta = \frac{\rho\, C_B\, (L/B)\, B^3}{(B/T)} \;\Rightarrow\; B = 0.8366\, \Delta^{1/3}$$

Student activity

Everything becomes a power of Δ

Q3 · Steel weight

Write the steel weight as a power of Δ and find its coefficients.

$$W_{\text{st}} = K\,E^{1.36} \qquad E = L\,(B+T) + 0.85\,L\,(D-T)$$

$$E = L^2\left[\frac{1}{(L/B)} + \frac{1}{(L/B)(B/T)} + \frac{0.85\,\big((D/T)-1\big)}{(L/B)(B/T)}\right] = 0.2509\,L^2$$

$$E = 0.2509\,(4.953)^2\,\Delta^{2/3} = c_E\,\Delta^{2/3}$$

$$W_{\text{st}} = K\,c_{\text{E}}^{1.36}\,\Delta^{\frac{2}{3}\times 1.36} = c_{\text{st}}\,\Delta^{n}$$

$$c_E = 6.156, \qquad c_{st} = 0.3789, \qquad n = 0.907$$

Q4 · Iteration function

Write g(Δ) with all coefficients evaluated, for fixed-point iteration

Equilibrium with the buoyancy in non-dimensional form and every weight as a power of Δ:

$$\underbrace{\rho\,C_{\text{B}}\,(L/B)\,(B/T)^2\,T^3}_{\Delta} = \text{DWT} + c_{\text{o}}\,\Delta + c_{\text{st}}\,\Delta^{n} + b\,\Delta^{2/3}$$

$$b = \frac{k_m\,V^3}{C_{adm}} = 0.6775$$

$$\Delta_{k+1} = g(\Delta_k) = \text{DWT} + c_{\text{o}}\,\Delta_k + c_{\text{st}}\,\Delta_k^{n} + b\,\Delta_k^{2/3}$$

Student activity

Q5 · Solve $\Delta = g(\Delta)$ by hand with fixed-point iteration: start at Δ_0 = DWT = 37 600 t and compute $\Delta_{k+1} = g(\Delta_k)$ with the $g(\Delta)$ of Q4, until the change $|\Delta_{k+1} - \Delta_k|$ is below 1 t. Then give Δ^*, the dimensions L, B, T, D and the weights: steel, outfit, machinery and lightship.

46 022.6 t in six evaluations of g

$$\Delta_{k+1} = \underbrace{37\,600}_{\text{DWT}} + \underbrace{0.3789\,\Delta_k^{0.907}}_{\text{steel}} + \underbrace{0.025\,\Delta_k}_{\text{outfit}} + \underbrace{0.6775\,\Delta_k^{2/3}}_{\text{machinery}}$$

Coefficients used: L/B = 5.92 · B/T = 2.69 · D/T = 1.36 · block coefficient = 0.757 · ρ = 1.025 t/m³

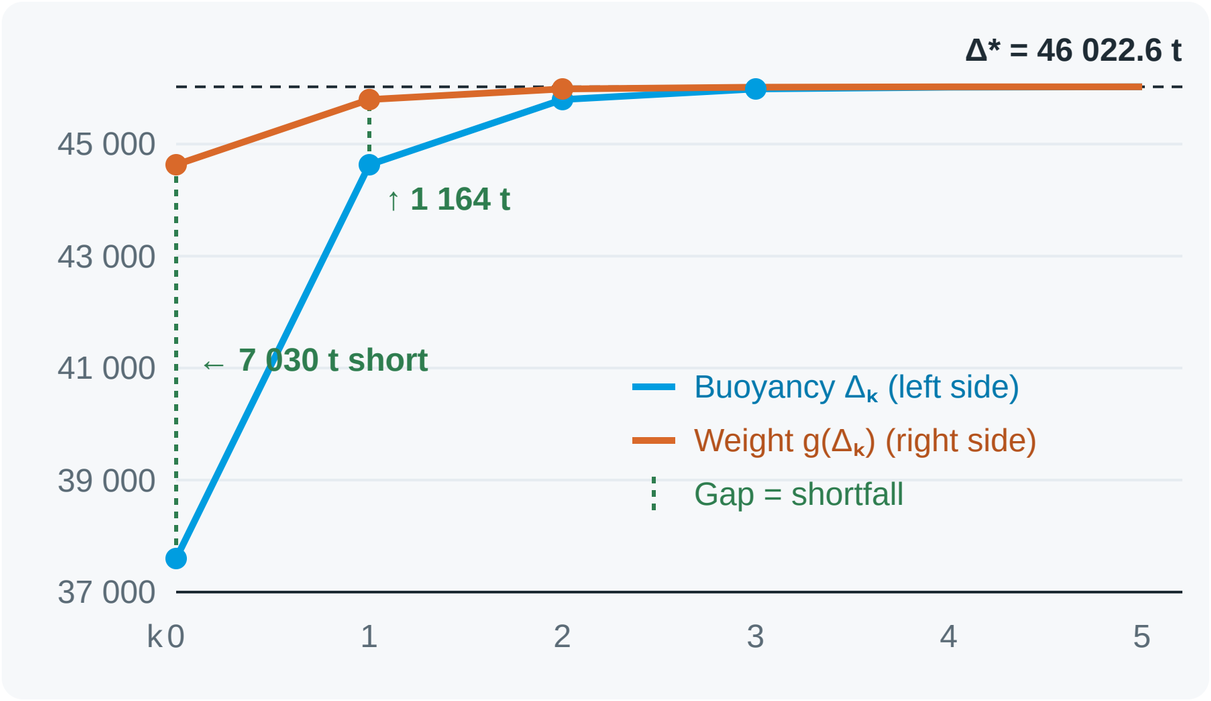
k	Δ_k (t)	T (m)	L (m)	E (m²)	Steel	Outfit	Machinery	Lightship	Δ_{k+1} (t)	Change (t)
	given	0.3110·Δ^⅓	(L/B)(B/T)·T	0.2509·L²	0.032·E^1.36	0.025·Δ	0.6775·Δ^⅔	Σ weights	DWT + Σ	$\Delta_{k+1} - \Delta_k$
0	37 600.0	10.420	165.93	6 909	5 330.0	940.0	760.4	**7 030.3**	44 630.3	7 030.3
1	44 630.3	11.032	175.69	7 745	6 226.2	1 115.8	852.4	**8 194.3**	45 794.3	1 164.0
2	45 794.3	11.127	177.20	7 879	6 373.2	1 144.9	867.2	**8 385.2**	45 985.2	190.9
3	45 985.2	11.143	177.45	7 901	6 397.3	1 149.6	869.6	**8 416.5**	46 016.5	31.3
4	46 016.5	11.145	177.49	7 905	6 401.2	1 150.4	870.0	**8 421.6**	46 021.6	5.1
5	46 021.6	11.146	177.49	7 905	6 401.9	1 150.5	870.0	**8 422.4**	46 022.4	0.8 < 1 ✓

k = 0 → 5: Δ grows +8 422 t but the lightship only +1 392 t (ratio 0.165 ≈ g′) → next slide. Weights in t.

Student activity

Q5 · continued: both sides of the equation, step by step

The lightship lags behind: +0.16 t per tonne



Step	Δ grows	Lightship grows	Ratio
0 → 1	+7 030.3	**+1 164.0**	**0.166**
1 → 2	+1 164.0	**+190.9**	**0.164**
2 → 3	+190.9	**+31.3**	**0.164**
3 → 4	+31.3	**+5.1**	**0.164**
4 → 5	+5.1	**+0.8**	**0.164**
0 → 5	**+8 421.6**	**+1 392.1**	**0.165**

$$\text{ratio} = \frac{LS(\Delta_k) - LS(\Delta_{k-1})}{\Delta_k - \Delta_{k-1}} \;\to\; g'(\Delta^*) = 0.164$$

The ratio is the **average slope of g over one step**, equal to g′ at a point inside it. Question 6 (extended) proves g′ ≤ 0.167, so each shortfall is at most 17% of the previous one. Weights in t.

Extended question

Q6 (extended) · Without iterating, prove that the iteration converges for any Δ ≥ DWT, and interpret g′(Δ).

Why it converges: g′ stays below 1

Criterion	$x_{k+1} = g(x_k)$ converges to x^* if $\lvert g'(x)\rvert < 1$ in a region containing x^*
Our problem	$x = \Delta, \quad \Delta^* > \mathrm{DWT} \Rightarrow$ show $\lvert g'(\Delta)\rvert < 1$ for all $\Delta \geq \mathrm{DWT}$

$$\Delta_{k+1} = g(\Delta_k) = \mathrm{DWT} + c_o\,\Delta_k + c_{st}\,\Delta_k^{\,n} + b\,\Delta_k^{\,2/3}$$

Differentiate term by term :

$$g'(\Delta) = c_o + n\,c_{st}\,\Delta^{n-1} + \tfrac{2}{3}\,b\,\Delta^{-1/3} = 0.025 + 0.3436\,\Delta^{-0.093} + 0.4516\,\Delta^{-1/3}$$

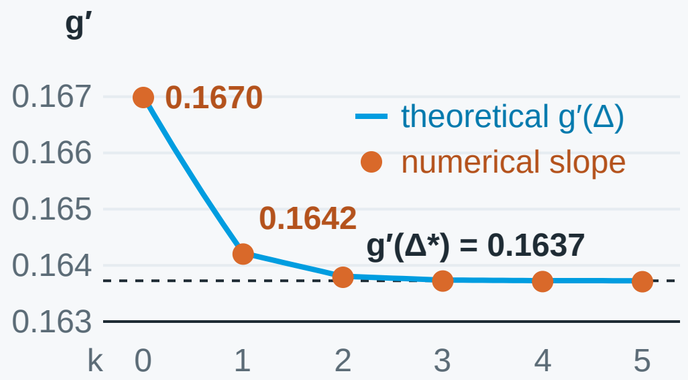
All terms are positive and decrease with Δ, so the maximum is at Δ = DWT:

$$g'(\mathrm{DWT}) = 0.025 + 0.3436 \times 37\,600^{-0.093} + 0.4516 \times 37\,600^{-1/3}$$
$$g'(\Delta) \leq g'(\mathrm{DWT}) = 0.025 + 0.1285 + 0.0135 = 0.167$$

Physical meaning: each extra tonne of displacement costs only 0.17 t of its own lightship, leaving **0.83 t** to float the shortfall.

With g′ > 1, every enlargement would sink the ship deeper.

k	Δ_k	$g(\Delta_k)$	$g(\Delta_k+100)$	slope
0	37 600.0	44 630.3	44 647.0	0.1670
1	44 630.3	45 794.3	45 810.7	0.1642
2	45 794.3	45 985.2	46 001.6	0.1638
3	45 985.2	46 016.5	46 032.9	0.1637
4	46 016.5	46 021.6	46 038.0	0.1637
5	46 021.6	46 022.4	46 038.8	0.1637



Three Things to Take Away

The design equation turns buoyancy = weight into one dimension, once the ratios and CB come from similar ships.

The weights become powers of Δ, so weight = buoyancy is one equation in Δ:

$$\Delta = \mathrm{DWT} + \sum_i c_i \, \Delta^{n_i}$$

Iterate from Δ_0 = DWT; it converges because the lightship grows more slowly than the ship.

Next: Wed 11 Nov, human factors in ship design (guest lecturer). Fri 13 Nov (W2.3): length, beam, depth, draught and form coefficients. Read §2.3–2.12.